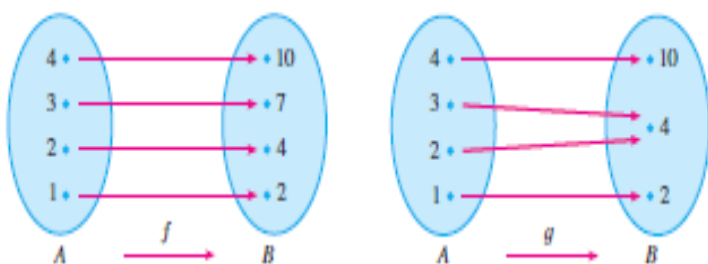


Unit 1.8 Inverse functions p 58 - 62

Note Title

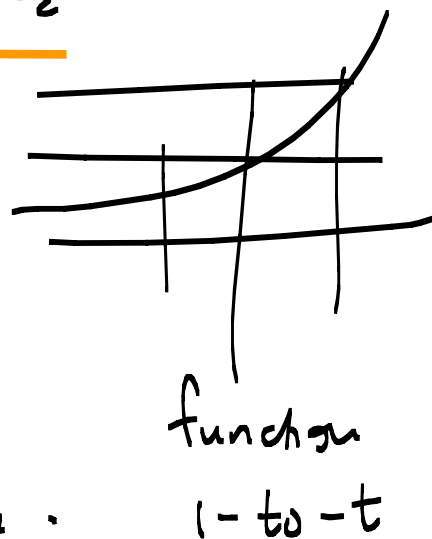
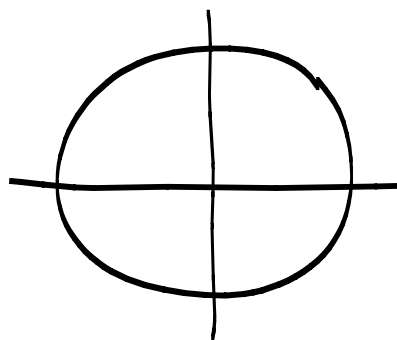
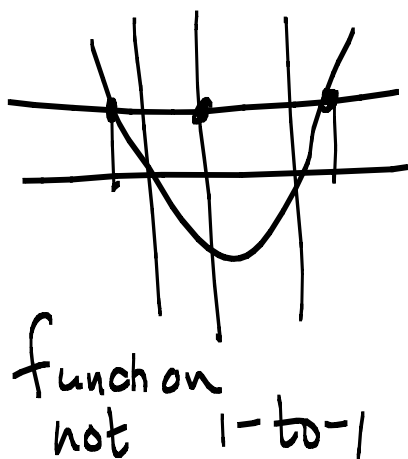
2014/02/08

One-to-one function never takes on the same value twice.



$f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ OR

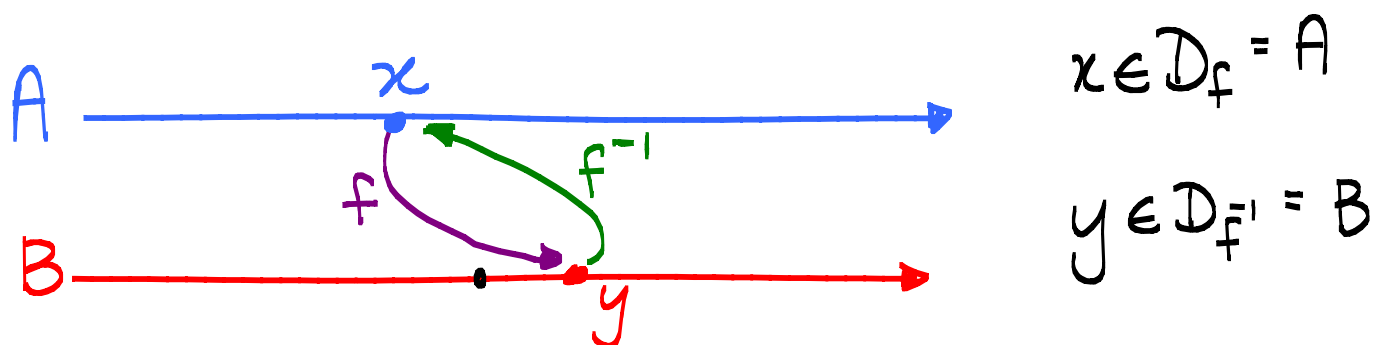
* f is a one-to-one function if
 $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$



2 Definition Let f be a one-to-one function with domain A and range B . Then its **inverse function** f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y$$

for any y in B .



Domain^(B) of f^{-1} = range^(B) of f
 Range^(A) of f^{-1} = domain^(A) of f

Cancellation equations

$$(f^{-1} \circ f)(x) = f^{-1}(f(\underline{x})) = x$$

$$(f \circ f^{-1})(x) = f(f^{-1}(x)) = x$$

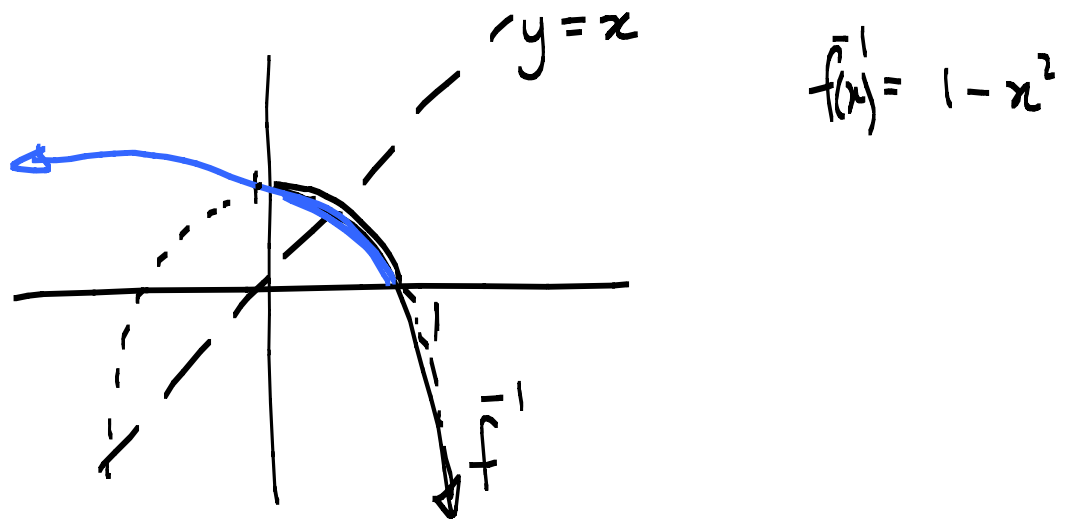
Find the inverse function of

$$\begin{matrix} 1-x \geq 0 \\ x \leq 1 \end{matrix} \quad \underline{f(x)} = \sqrt{1-x} \quad f^{-1}(x) = 1-x^2$$

$$\textcircled{1} \quad y = \sqrt{1-x} \quad D_f = (-\infty, 1] \quad R_{f^{-1}} = \underline{\underline{(-\infty, 1]}}$$

$$\textcircled{2} \quad x = \sqrt{1-y} \quad R_f = [0, \infty) \quad \underline{\underline{D_{f^{-1}} = [0, \infty)}}$$

$$\textcircled{3} \quad x^2 = 1-y \Rightarrow y = 1-x^2$$



f^{-1} - notation "the inverse function of f ".

~~$f^{-1} = \frac{1}{f}$~~

Sketch $f(x) = \sqrt{10 - 3x}$ and find its inverse as well as D_f and R_f and $D_{f^{-1}}$ and $R_{f^{-1}}$.

① let $y = \sqrt{10 - 3x}$

② $y^2 = 10 - 3x \Rightarrow x = \frac{10}{3} - \frac{y^2}{3}$

③ let $y = \frac{10}{3} - \frac{x^2}{3}$ $D_f = (-\infty, \frac{10}{3}]$

$\Rightarrow f^{-1}(x) = \frac{10}{3} - \frac{x^2}{3}$ $R_{f^{-1}} = (-\infty, \frac{10}{3}]$

$R_f = [0, \infty)$

$D_{f^{-1}} = [0, \infty)$

Sketch
SELF.